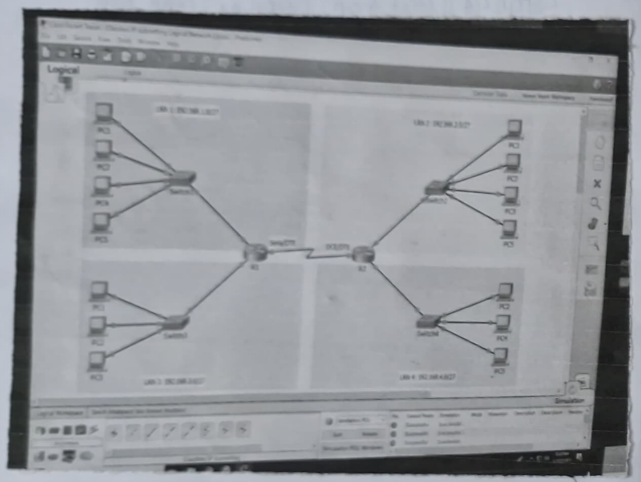


EXP-9 · Implementation of subnetting in cisco packet tracer

- **Aim:**

  To implement classless IP subnetting using the cisco packet tracer simulator, configure routers switches and PCs and analyze packet transmissions.

- **Theory:**

  - subnetting: the technique of logically dividing a single large network into smaller subnetworks
  - classless subnetting: Allows custom subnet masks rather than standard default classful boundaries (class A, B, C) to optimize address
  - subnet calculation.

    network address: 192.168.10.0/24

    subnet mask: (255.255.255.224)

    Borrowed bits: $2^3 = 8$ subnets

- **Procedure:**

  ① Step 1 - Topology design:

  Open cisco tracer and place 1 router.

  Connect PCs to their respective switches using copper cable

  ② Step ② Router interface configuration.

interface fastethernet 0/1
ip address 192.168.2.1
no shut down
exit

step ③ - switch port configuration
enable, configure terminal; switchport mode access
exit

step ④ Host configuration
click each PC, assign static IP, subnet mask and default gateway in the fast ipar

Result:
Classless IP subnetting was implemented and tested using Cisco packet tracer. Communication between host d/w the subnet through a router was verified when fully